

Lecture 1: Calculus I Integration Review

Monday, August 24 • Sections 1.4–1.7

• Guided Notes •

How to Use These Guided Notes: These guided notes are designed to help you organize the main ideas from each lecture but

1. we will not complete every example in class.
2. Although most definitions and theorems are included, you may need to add additional notes, examples, or explanations during class.
3. Occasionally, you will see sections labeled **Deeper Insight**. These sections provide interesting background information and help develop a deeper understanding of calculus, but they are not part of the core Calculus II content.

Before reviewing integration techniques, let us briefly refresh the differentiation rules that will be needed throughout this course.

Remember that all these denote the derivative of a function f : $f'(x)$, $\frac{df}{dx}$, $\frac{d}{dx}f(x)$.

Try it yourself.

1. $\frac{d}{dx}(2)$.

4. $\frac{d}{dx}(\sqrt[3]{x^2})$.

7. $\frac{d}{dx}\left(\frac{\sin x}{1 + \cos x}\right)$.

2. $\frac{d}{dx}(x^{\frac{3}{4}})$.

5. $\frac{d}{dx}(e^{2x} - 1)$.

8. $\frac{d}{dx}(5^{x^2})$.

3. $\frac{d}{dx}\left(\frac{1}{x^2}\right)$.

6. $\frac{d}{dx}(x \ln(x))$.

9. $\frac{d}{dx}\left(\frac{1}{\sqrt[3]{x^2 + x + 1}}\right)$.

10. Determine $f'(x)$ if

(a) $f(x) = \arccos(\sqrt{x})$.

(c) $f(x) = (\tan^{-1}(x))^2$.

(b) $f(x) = \sec^{-1}(\sqrt{x-1})$.

(d) $f(x) = \sin^{-1}(5x)$.

Before reviewing the substitution rule, let us recall several key concepts from Calculus I.

Definition 1.1 – Antiderivative

A function F is an *antiderivative* of f on an interval I if

$$F'(x) = \underline{\hspace{2cm}} \quad \text{for all } x \text{ in } I.$$

Understanding the definition

1. Which of the following functions are antiderivatives of $\sin(2x)$?

- (a) $\frac{1}{2} \cos(x)$ (b) $-2 \cos(x)$ (c) $2 \cos(x) + 1$ (d) $1 - \frac{1}{2} \cos(x)$

2. Is this statement true or false: *If $F(x)$ is an antiderivative of $f(x)$, then for every real number C , the function $H(x) = F(x) + C$ is also an antiderivative of $f(x)$.* Why?

3. Is this statement true or false: *Every function f has infinitely many antiderivatives.* Why?

Deeper Insight – Functions with no or complicated antiderivatives

Functions with jump discontinuities cannot be the derivative of another function, which means that functions with jump discontinuities do not have antiderivatives. As a consequence, the second statement above is false. For example, the function

$$f(x) = \begin{cases} 1 & \text{if } x \geq 0, \\ -1 & \text{if } x < 0 \end{cases}$$

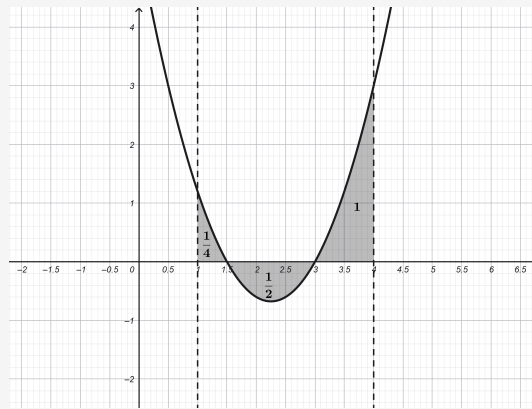
does not have any antiderivative.

Other functions, like e^{-x^2} have antiderivatives, but they are not elementary functions (they cannot be written in terms of the functions we are used to work with, like polynomials, roots, trigonometric functions, logarithmic or exponential functions). At the end of this course, we will see how to deal with this type of integrals.

Definition 1.2 – Definite Integral

If a and b are real numbers and $f(x)$ is a function, integrable over $[a, b]$, the definite integral $\int_a^b f(x) dx$ is a real number, the **signed** area of the region between the graph of the function, the x -axis and the lines $x = a$ and $x = b$. It is the limit of an infinite sum, called a Riemann sum. If f is a continuous function on $[a, b]$, then f is integrable on $[a, b]$, and the definite integral $\int_a^b f(x) dx$ exists.

What does this mean? The definite integral $\int_a^b f(x) dx$ is not the area of the region enclosed by the graph of f , the x -axis and the vertical lines $x = a$ and $x = b$, in some cases it is, but in general, it is a signed area, it may be positive or negative. If the function $f(x)$ is as represented in this figure (the numbers $\frac{1}{4}$, $\frac{1}{2}$, and 1 are the areas of the corresponding regions):



determine

$$\int_1^{1.5} f(x) dx = \underline{\hspace{2cm}}, \quad \int_{1.5}^3 f(x) dx = \underline{\hspace{2cm}}, \quad \int_3^4 f(x) dx = \underline{\hspace{2cm}}.$$

If $f(x) < 0$ somewhere on $[a, b]$, $\int_a^b f(x) dx$ is not equal the *geometric* area of the region between the curve, the x -axis, and the lines $x = a$ and $x = b$.

The Fundamental Theorem of Calculus relates two previously unrelated ideas: the signed area under a curve and antiderivatives. It has two parts. In this course, we will mainly use Part II, which allows us to evaluate definite integrals using antiderivatives. Part I is included because it reveals the deeper connection between accumulation, area, and differentiation, one of the central ideas of calculus.

Deeper Insight – Fundamental Theorem of Calculus, Part 1

If f is continuous on $[a, b]$, the function $g(x)$ defined by $g(x) = \int_a^x f(t) dt$, satisfies $g'(x) = f(x)$, i.e. $g(x)$ is one antiderivative of $f(x)$.

Theorem 1.3 – Fundamental Theorem of Calculus, Part 2

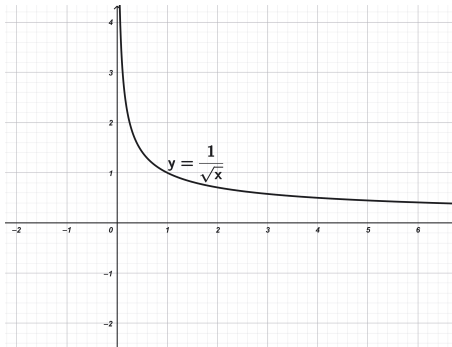
If a and b are real numbers, $f(x)$ is continuous on $[a, b]$ and $F(x)$ is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(x) dx = \underline{\hspace{2cm}}.$$

Using the theorem. Evaluate $\int_1^4 \frac{1}{\sqrt{x}} dx$.

Note that the function $\frac{1}{\sqrt{x}}$ is continuous on $[1, 4]$. **We may use the Fundamental Theorem of Calculus to evaluate the definite integral**, all we need is to determine one antiderivative of $f(x) = \frac{1}{\sqrt{x}}$.

1. Write one antiderivative of $f(x)$: $\underline{\hspace{2cm}}$.
2. Evaluate it, the definite integral is equal to: $\underline{\hspace{2cm}}$.
3. Shadow the region whose area you determined above:



Let us take a more careful look at the second part of the Fundamental Theorem of Calculus.

What does this mean? Two important assumptions for the Fundamental Theorem of Calculus, part 2, are that a and b are real numbers and f is continuous on $[a, b]$. Cross all the integrals where the Fundamental Theorem of Calculus **cannot** be applied?

☐ $\int_{-1}^1 \frac{1}{x} dx.$

☐ $\int_1^2 \frac{1}{x} dx.$

☐ $\int_0^{\frac{\pi}{2}} \tan(x) dx.$

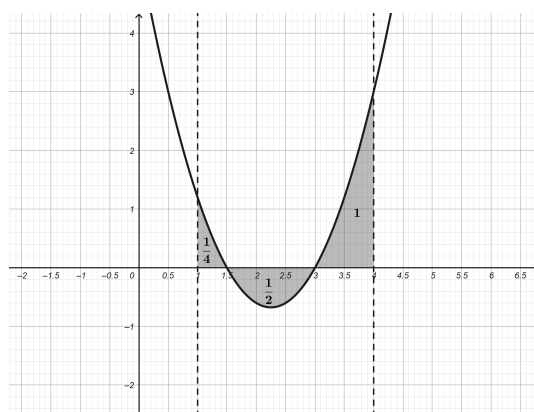
☐ $\int_0^2 \sqrt{x} dx.$

Some properties of the definite integral are the following, assuming that a, b are real numbers and f is continuous on $[a, b]$,

1. $\int_a^a f(x) dx = 0.$
2. $\int_a^b f(x) dx = -\int_b^a f(x) dx.$

3. For any c between a and b , $\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx$.
4. $\int_a^b (f(x) \pm g(x)) \, dx = \int_a^b f(x) \, dx \pm \int_a^b g(x) \, dx$.
5. For any real number c , $\int_a^b cf(x) \, dx = c \int_a^b f(x) \, dx$.

Try it yourself. If the function $f(x)$ is as represented in this figure (the numbers $\frac{1}{4}$, $\frac{1}{2}$, and 1 are the areas of the corresponding regions):



Use the properties of the definite integral to determine

$$\int_1^4 f(x) \, dx = \underline{\hspace{2cm}}, \quad \int_1^4 2f(x) \, dx = \underline{\hspace{2cm}}.$$

Definition 1.4 – Indefinite Integral

Because of this relation between antiderivatives and definite integrals (areas) given by the Fundamental Theorem of Calculus, if a function $f(x)$ has antiderivatives, the symbol $\int f(x) \, dx$ (indefinite integral of $f(x)$) is used to denote its family of antiderivatives, i.e., if $F(x)$ is one antiderivative of $f(x)$, the indefinite integral of $f(x)$ is

$$\int f(x) \, dx = F(x) + C.$$

Due to the differentiation rules for a sum/difference of functions and the product of a function with a constant c , it holds

1. $\int (f(x) \pm g(x)) \, dx = \int f(x) \, dx \pm \int g(x) \, dx$.
2. For any real number c , $\int cf(x) \, dx = c \int f(x) \, dx$.

! Common Mistakes

Remember, $\int f(x) dx$ and $\int_a^b f(x) dx$ have different meanings:

- $\int f(x) dx$ is the family of antiderivatives of f . When solving $\int f(x) dx$ we are looking for all the functions whose derivative equals $f(x)$.
- $\int_a^b f(x) dx$ is a signed area. When solving $\int_a^b f(x) dx$, we are calculating a signed area.

Two mistakes we may do if we forget this are the following:

- Writing “ $+C$ ” on a *definite* integral or **not** writing “ $+C$ ” on an *indefinite* integral.
- Assuming $\int_a^b f(x) dx$ is automatically “the area,” even when f is negative on part of $[a, b]$.

In the next three lectures, we will learn how to use integration to find areas of planar regions and volumes of three-dimensional solids. To accomplish this, we need to know how to evaluate definite integrals. In many cases, the integrand is not immediately recognizable as the derivative of a known function, so we must use techniques for finding antiderivatives. One such technique, which you learned in Calculus I, is **the substitution rule**.

Theorem 1.5 – The Substitution Rule

1. Substitution rule for indefinite integrals $\int f(g(x))g'(x) dx$: applying the substitution

$$u = g(x), \quad du = \underline{\hspace{2cm}},$$

$$\int f(g(x)) g'(x) dx = \underline{\hspace{2cm}}.$$

2. Substitution rule for definite integrals $\int_a^b f(g(x))g'(x) dx$: applying the substitution

$$u = g(x), \quad du = \underline{\hspace{2cm}},$$

$$\int_a^b f(g(x)) g'(x) dx = \underline{\hspace{2cm}}.$$

When do I use this? When the function to integrate is a composite function, look for an *inner* function whose derivative (up to a constant multiple) also appears in the integrand.

Using the theorem. Let us calculate an indefinite integral: $\int x \cos(x^2) dx$.

1. Choose u : $u =$ _____.
2. Compute du : $du =$ _____.
3. Rewrite the integral in u : $\int$ _____ du .
4. Integrate, then substitute back: _____.

Using the theorem. Let us calculate a definite integral: $\int_0^{2\pi} \sin x \cos x dx$.

1. Choose u : $u =$ _____.
2. Compute du : $du =$ _____.
3. Rewrite the integral in u : all elements of the new definite integral **must** be u -elements (the limits of integration, the integrand, the variable of integration)

$$\int_{\boxed{}}^{\boxed{}} \boxed{} du.$$

4. Integrate and evaluate: _____.

Using the theorem. Let us calculate a definite integral: $\int_4^{16} \frac{e^{-\sqrt{x}}}{\sqrt{2x}} dx$.

1. Choose u : $u =$ _____.
2. Compute du : $du =$ _____.
3. Rewrite the integral in u : all elements of the new definite integral **must** be u -elements (the limits of integration, the integrand, the variable of integration)

$$\int_{\boxed{}}^{\boxed{}} \boxed{} du.$$

4. Integrate and evaluate: _____.

When applying u -substitution to definite integrals, a consistent treatment of the limits of integration is important. Two specific errors frequently occur during this process.

! Common Mistakes

Mistake 1: Failure to Transform Limits of Integration when applying the substitution rule to a definite integral.

The substitution $u = g(x)$ and $du = g'(x)dx$ is applied correctly to the integrand but the original x -limits are retained for the u -integral.

Correction: The limits of integration must be updated using the substitution formula $u = g(x)$ so that the integral is evaluated over the correct interval in the u -domain.

For example, suppose we want to evaluate

$$\int_0^1 2x(x^2 + 1)^3 dx. \quad (1)$$

After applying the substitution $u = x^2 + 1$, $du = 2xdx$, writing the new definite integral as $\int_0^1 u^3 du$ is **INCORRECT**. Instead, you have to update the limits: $u(0) = 0^2 + 1 = 1$ and $u(1) = 1^2 + 1 = 2$ and write the new definite integral as $\int_1^2 u^3 du$.

Mistake 2: Inconsistent Variable Reversion

In this case, the limits are correctly transformed to u -values, but the antiderivative is converted back into terms of the original variable x before evaluation. This results in plugging u -limits into an x -expression.

Correction: Once the limits are transformed to u , the integral should be evaluated entirely in the u -domain. Alternatively, if the variable is reverted to x , the original x -limits must be restored. Mixing u -limits with x -variables is notationally and mathematically invalid.

For example, suppose we want to evaluate (1). Applying the substitution proposed above, the new correct definite integral is

$$\int_1^2 u^3 du = \frac{1}{4} u^4 \Big|_1^2.$$

Writing the result in terms of the original variable without changing back the limits of integration is **INCORRECT**, continuing like this

$$\frac{1}{4} u^4 \Big|_1^2 = \frac{1}{4} (x^2 + 1)^4 \Big|_1^2 = \frac{1}{4} (5^4 - 2^4)$$

is **INCORRECT**. Instead, one should evaluate directly in u using the new limits

$$\frac{1}{4} u^4 \Big|_1^2 = 4 - \frac{1}{4} = \frac{15}{4}.$$

In summary, for a definite integral $\int_a^b f(g(x))g'(x)dx$, the substitution $u = g(x)$ requires the transformation to $\int_{g(a)}^{g(b)} f(u)du$. Evaluation should proceed with the new limits and the new variable

without returning to the original variable.

Practice Problems

Level I Problems

- | | | |
|--|---|--|
| 1. $\int_0^1 (x^e + e^x) \, dx.$ | 5. $\int \frac{x}{\sqrt{4-x^2}} \, dx.$ | 9. $\int \frac{x^3}{\sqrt{9+x^4}} \, dx.$ |
| 2. $\int_1^{18} \sqrt{\frac{3}{z}} \, dz.$ | 6. $\int \frac{x^2}{x^3+1} \, dx.$ | 10. $\int 3x^2 (x^3-3)^2 \, dx.$ |
| 3. $\int_{-1}^1 \frac{1}{x+2} \, dx.$ | 7. $\int_1^3 (2-x)^6 \, dx.$ | 11. $\int -4e^x (6e^x+6)^5 \, dx.$ |
| 4. $\int_{-1}^1 x\sqrt{9-x^2} \, dx.$ | 8. $\int_1^e \frac{\ln x}{x} \, dx.$ | 12. $\int \frac{1}{\sqrt{1-16x^2}} \, dx.$ |
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Practice Problems

Level II Problems

- | | |
|---|---|
| 1. $\int_0^{\frac{\pi}{6}} 5 \tan(2x) \ln(\sec(2x)) \, dx.$ | 8. $\int \frac{\sin(\tan^{-1}(t))}{1+t^2} \, dt.$ |
| 2. $\int_1^{\sqrt{2}} x 2^{x^2-1} \, dx.$ | 9. $\int_0^1 \frac{x^2+2x}{x^3+3x^2+1} \, dx.$ |
| 3. $\int_0^1 \frac{e^{2t}}{e^t+1} \, dt.$ | 10. $\int \frac{x^2}{\sqrt{x-1}} \, dx.$ |
| 4. $\int_0^2 \frac{1}{\sqrt{x}(2+\sqrt{x})^2} \, dx.$ | 11. $\int_{-8}^{-6} \frac{4x}{x+4} \, dx.$ |
| 5. $\int \frac{\sqrt{2+\sqrt{t}}}{\sqrt{t}} \, dt.$ | 12. $\int_0^{\sqrt{\pi}} t \cos(t^2) \sin(t^2) \, dt.$ |
| 6. $\int \frac{1+t}{1-t} \, dt.$ | 13. $\int (8-36x^2) \sec(3x^3-2x) \tan(3x^3-2x) \, dx.$ |
| 7. $\int_2^8 \frac{-2e^{\sqrt{2x}}}{\sqrt{2x}} \, dx.$ | 14. $\int (-5x^2) \sec^2(3x^3) \tan^3(3x^3) \, dx.$ |

Practice Problems

Level III Problems

1. $\int_0^{\frac{\pi}{4}} \frac{\sin^3(x)}{\cos^4(x)} dx$. **Hint:** $\sin^3(x) = \sin^2(x) \sin(x) = (1 - \cos^2(x)) \sin(x)$.
 2. $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos^3(\theta) d\theta$.
 3. $\int_{-6}^6 \frac{e^x}{1 + e^{2x}} dx$.
 4. $\int \frac{1}{x + x \ln^2 x} dx$.
 5. If $F(x)$ is one antiderivative of $f(x)$, evaluate $\int_1^4 f(-x) dx$ and $\int_1^4 f(2x) dx$.
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Checking Your Understanding

1. True or False: $\int \frac{\log_{10}(x)}{x} dx = \frac{1}{2} \log_{10}^2(x)$.
 2. Let $f(x)$ be a continuous function. Suppose one student computes the indefinite integral $\int f(x) dx$ and obtains the result $F(x) + 1 + C$. Using *Wolfram Alpha*, the same integral is shown as $F(x) + C$. Knowing that *Wolfram Alpha*'s result is correct, cross the true statement.

☐ The results are different. The student's answer is wrong.

☐ The results are different, but the student's answer is correct.
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After this lesson I can:

- ☐ Differentiate common algebraic, exponential, logarithmic, trigonometric, and inverse trigonometric functions.
 - ☐ Explain what an antiderivative is.
 - ☐ Explain why antiderivatives differ by a constant.
 - ☐ Interpret a definite integral as a signed area.
 - ☐ Apply the Fundamental Theorem of Calculus

- to evaluate a definite integral.
 - ☐ Distinguish between a definite integral and an indefinite integral.
 - ☐ Use the substitution rule to evaluate indefinite integrals.
 - ☐ Use the substitution rule to evaluate definite integrals, including changing the limits of integration.

Muddiest Point: What concept from today's lesson is still unclear?